

INTERACTIVE INDOOR WAYFINDING AND NAVIGATION KIOSK SYSTEM

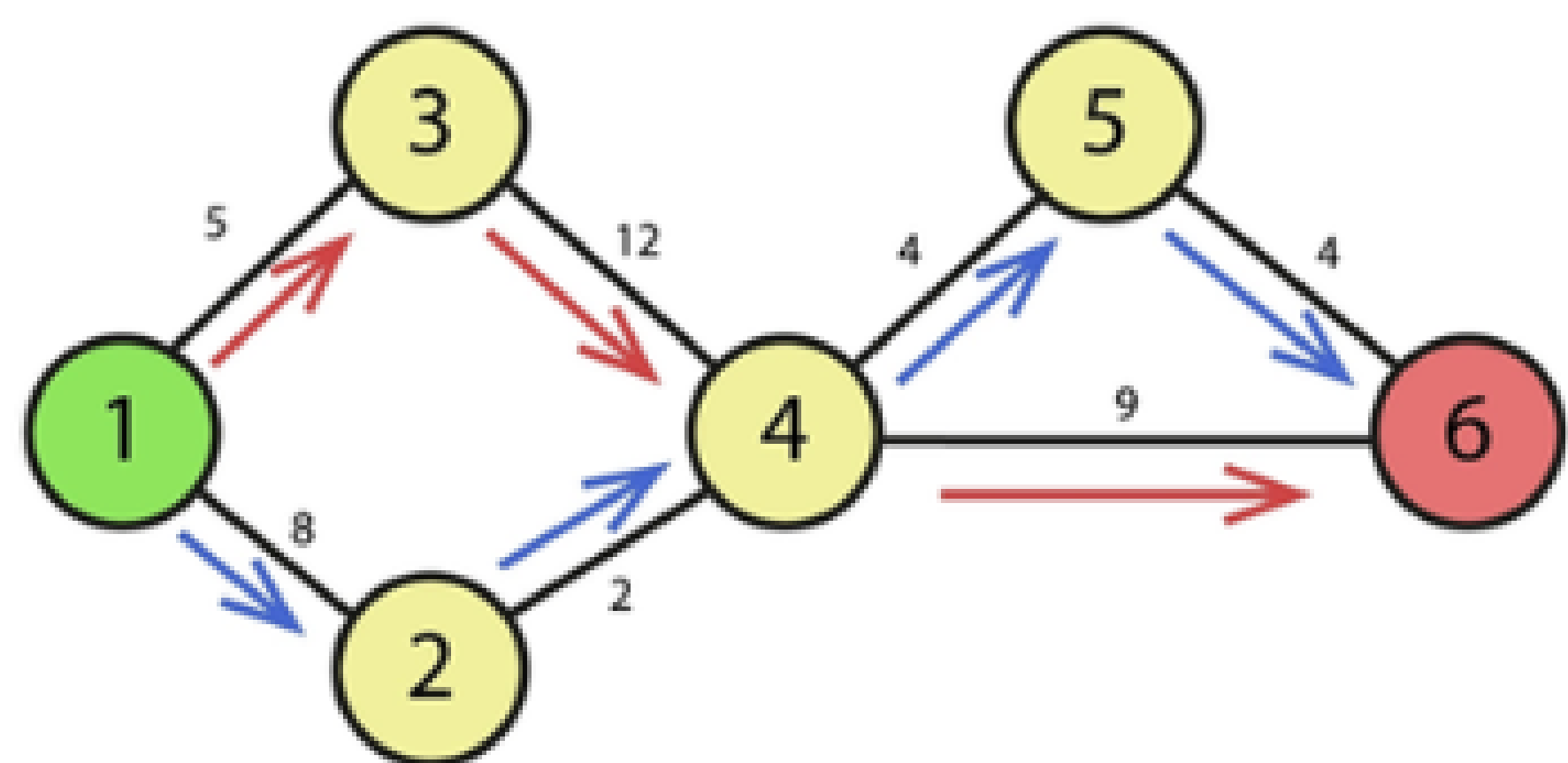
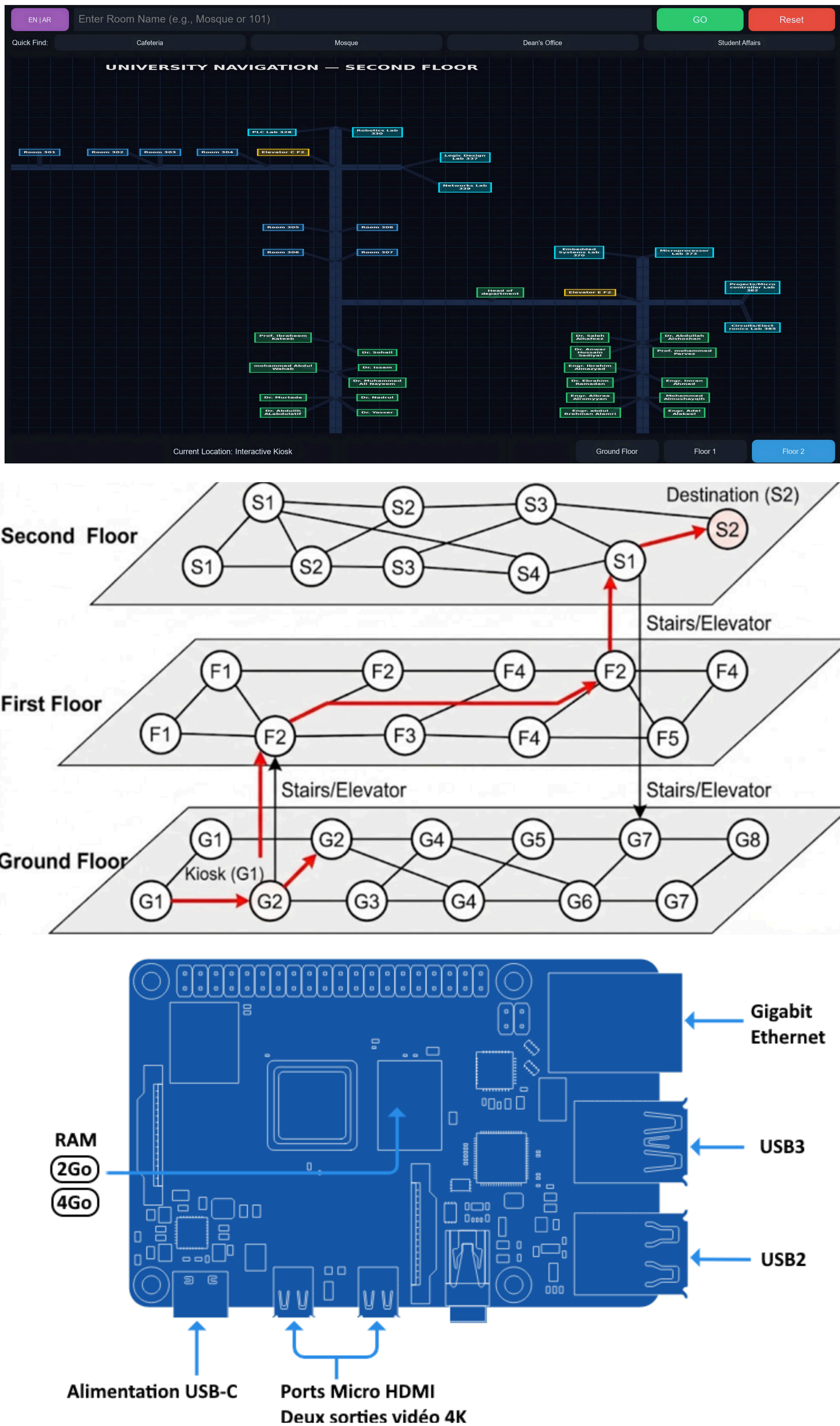
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INTRODUCTION

- *Large-scale environments suffer from the "Indoor Wayfinding Paradox," where complex layouts make navigation difficult and traditional static maps become quickly outdated.*
- *The Solution: This project introduces an "Interactive Indoor Wayfinding Guidance System" to bridge the gap between complex physical layouts and user-friendly navigation.*
- *Application: The system provides dynamic, visual, multi-floor routing to assist new visitors.*

METHODOLOGY

- *Graph Model: Floor plan mapped as a weighted, undirected graph.*
- *Pathfinding: Dijkstra's Algorithm calculates optimal multi-floor routes via elevators.*
- *Hardware & Data: Raspberry Pi 4 Model B (Linux OS) with localized SQLite database.*



Dijkstra’s Algorithm

GOALS AND OBJECTIVES

- 🌐 **Universal Engine:**Adaptable Dijkstra’s Algorithm.
- 🗄️ **Central Data:** Robust local database for coordinates.
- 💻 **Cost-Effective:** Utilizes efficient Raspberry Pi SBCs over expensive industrial PCs.
- 📍 **Accessibility:** Users find their destination in less than 10 sec.



CONCLUSION

- *Vision 2030: Supports national Digital Transformation and "Smart Building" technology*
- *Commercial Viability: Highly scalable and cost-competitive against traditional signage.*
- *Environmental Impact: Eliminates physical map e-waste.*